| 평가 항목 | 배점 |
|---|---|
| (1) $\alpha = \frac{2\pi}{17}$ | 0,2 |
| (2) $d = F\,\alpha$ | 0,6 |
| (3) $l = F + d\,tg\,\alpha \approx F(1+\alpha^2)$ | 0,4 |
| (4) $R = \frac{l}{2tg\frac{\alpha}{2}} \approx \frac{F(1+\alpha^2)}{\alpha}$ | 0,4 |
| $\alpha^2$ | *(-0,2)* |
| $R = 30{,}8\ sm$ | 0,2 |
| (5) $l = F - F\,tg^2\alpha \approx F\left(1-\alpha^2\right)$ | 0,6 |
| (6) $R \approx F\frac{1-\alpha^2}{\alpha}$ | 0,4 |
| $\alpha^2$ | *(-0,2)* |
| $R = 23{,}4\,sm$ | 0,2 |
| **합계** | **3,0** |

# 문제 2. 산의 물리학 (10,0점)

## 제1 부. 등온 대기 (3,2점)

1.1 **[1,0점]** ,

$$p_0 S = Mg, \tag{1}$$

$$S = 4\pi R_E^2 \tag{2}$$

.

(1) (2)

$$M = \frac{4\pi p_0 R_E^2}{g} = 5.32 \cdot 10^{18}\ \text{kg}. \tag{3}$$

1.2 **[1,0점]** .

$dh$ .

$dp$ $\rho$,

$$dp = -\rho g dh. \tag{4}$$

$$\rho = \frac{\mu_{air} p}{RT_0}. \tag{5}$$

(4) (5) $h$

$$p(h) = p_0 \exp\left(-\frac{\mu_{air} g}{RT_0} h\right) \tag{6}$$

$H = 1500\ m$

$$p(H) = 85.0 \cdot 10^3\ \text{Pa}. \tag{7}$$

1.3 **[0,6점]**

. ,

$$\delta Q = \frac{M}{\mu_{air}} \frac{\gamma R}{\gamma - 1} \Delta T = 5.33 \cdot 10^{21}\ \text{J}, \tag{8}$$

$$\gamma = 7/5. \tag{9}$$